

PROBABILISTIC ACTION PREDICTION OF HUMAN DRIVERS USING DYNAMIC BAYESIAN NETWORKS

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ABSTRACT

Autonomous vehicles (AVs) are expected to operate for many years in mixed traffic with human-driven vehicles (HDVs), where estimating human maneuvers is important for safe interaction. This paper presents a probabilistic framework for short-horizon human driver maneuver prediction using Dynamic Bayesian Networks (DBNs). The model is trained on the highD naturalistic highway dataset using window-based observations from vehicle motion and surrounding traffic interactions. The DBN uses a hierarchical latent structure, where action states are conditioned on broader driving regimes, allowing maneuver type and interaction context to be modeled together.

Two model variants are evaluated: a model without lane-change reweighting and a lane-change-weighted model. On the test set, the model without lane-change reweighting gives the best performance, with 79.4% exact next-step accuracy, 67.6% macro precision, 71.5% macro F1-score, and an 81.3% hit rate over a 2 s prediction horizon. The lane-change-weighted model reaches 64.2% exact accuracy and a 67.5% horizon hit rate. The average CPU latency for online belief update and next-maneuver prediction is 0.190 ms per vehicle, far below the 400 ms input interval. The results show that DBNs can support fast and interpretable short-horizon maneuver prediction.

Keywords: Human Driver Behavior Modeling, Autonomous Driving, Human-Driven Vehicles, Dynamic Bayesian Networks, Maneuver Prediction, Probabilistic Modeling, Mixed Traffic Environments

1. INTRODUCTION

Recent progress in artificial intelligence (AI), sensing, computing, and communication has accelerated the development of AVs. However, safe operation still depends on how well AVs can handle mixed traffic with HDVs, since fully autonomous traffic systems do not yet exist.

In mixed traffic, an AV cannot directly observe the intentions of nearby HDVs. Instead, human driver behavior must be inferred

from observable cues such as vehicle motion, lane changes, relative positions, and interactions with other nearby vehicles [1]. Human driver behavior prediction is therefore an important problem in autonomous driving. Estimating likely future maneuvers can support downstream tasks such as risk assessment, trajectory prediction, and motion planning [1–3].

Existing behavior prediction methods include rule-based, probabilistic, and learning-based approaches [2, 4, 5]. Rule-based models are simple and interpretable but often too rigid to capture the variability of real human behavior. Learning-based methods can learn complex patterns from large datasets, but many are difficult to interpret and are black boxes. Probabilistic models provide a useful middle ground because they can represent uncertainty, capture temporal dependencies, and remain interpretable.

This paper focuses on probabilistic modeling of high-level human driving behavior in highway traffic, with the goal of predicting likely future maneuvers of surrounding drivers while explicitly representing uncertainty. In the proposed framework, future maneuvers are represented as discrete action states, and a DBN models the temporal relationship between trajectory-based observations and hidden behavioral states. Driver intent is represented through latent variables and inferred probabilistically from observed vehicle motion and traffic context [6]. The model is trained and evaluated on the highD dataset [7] using features that describe vehicle motion, surrounding traffic context, and interaction behavior.

The study evaluates whether this framework can provide accurate, interpretable, and computationally efficient short-horizon predictions of future human driving maneuvers.

The main contributions of this paper are threefold: (1) a structured DBN framework for probabilistic short-horizon maneuver prediction with interpretable latent states, (2) a window-based observation design that combines ego motion with surrounding traffic interaction features, and (3) an evaluation on the highD dataset covering predictive accuracy, short-horizon hit rate, and online inference latency.

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2. RELATED WORK

Human driver behavior and maneuver prediction methods are commonly grouped into rule-based, probabilistic, and learning-based approaches, which differ in how they represent driver decisions, uncertainty, and temporal dependencies.

2.1. Rule-Based Driver Models

Rule-based models describe driving behavior through explicit mathematical equations or logical rules. Typical examples include car-following and lane-change models such as the Intelligent Driver Model (IDM), Gipps' model and the Minimizing Overall Braking Induced by Lane Changes (MOBIL) model, which relate observable traffic variables to longitudinal or lateral driving actions [8–11].

Their main strengths are interpretability, low computational cost, and clear physical meaning of parameters. However, they usually rely on hand-crafted assumptions and often represent driver behavior in a simplified or mostly deterministic way. This limits their ability to capture uncertainty and variability across drivers in real traffic.

2.2. Probabilistic Temporal Models

Probabilistic models address uncertainty by describing driver behavior as a stochastic process. Instead of predicting a single fixed response, they represent multiple possible behaviors and quantify uncertainty over future actions [12]. This is useful for human-driver prediction because the same observed traffic situation can lead to different driver responses.

Among probabilistic methods, Hidden Markov Models (HMMs) are a common starting point for modeling time-varying driving behavior. They represent behavior as a sequence of latent states that generate observable signals [13]. In driver modeling, these hidden states can correspond to maneuver-related modes such as lane keeping, braking, or overtaking [14]. HMMs are attractive because of their simple structure and efficient inference. However, they represent the hidden state with a single variable, which limits how much structure can be encoded in the model.

DBNs extend this temporal modeling idea by representing the state of a system with several variables across time [6, 15]. This makes it possible to encode temporal dependencies between hidden variables, observations, and contextual factors in a structured probabilistic model. In driver-behavior research, Bayesian-network-based models have been used for maneuver prediction, lane-change reasoning, and safety-related inference [16–18].

Li et al. proposed a DBN-based lane-change maneuver prediction framework for highway driving using historical vehicle states, road structure, and traffic interaction features. Their model achieved an F1-score of 80% for lane-change recognition and an advance prediction time of 3.75 s [16]. Talluri et al. used a DBN to combine lateral evidence with safety assessment for cut-in and lane-change reasoning. Mlynarczyk et al. focused on an affective driver-behavior module, where Bayesian networks estimate mental states such as mental load and fatigue from physiological and demographic variables. In contrast, the present work learns a compact DBN directly from naturalistic highD trajectory windows, using a factored latent structure with style and action

variables. The learned joint latent states are mapped to semantic maneuver labels only after training, allowing short-horizon maneuver prediction while keeping the latent behavior representation interpretable.

Other probabilistic approaches, such as Gaussian Mixture Models (GMMs), Gaussian Mixture Regression (GMR), and Gaussian Process Regression (GPR), are also used in driving-behavior and trajectory modeling [19–21]. These methods are useful for continuous quantities such as acceleration or trajectory evolution, but the present paper focuses on discrete short-horizon maneuver prediction with a structured temporal model.

2.3. Learning-Based Prediction Models

Learning-based methods form another major family of behavior prediction models. Classical machine learning methods such as Support Vector Machine (SVM), decision trees, and random forests were early data-driven approaches for intent recognition and maneuver classification [22]. Recent data-driven models have reported high performance for lane-change-specific tasks on highway trajectory datasets. For example, Xue et al. reported lane-change decision accuracy increasing from 97.02% to 98.20% when traffic context was included, while Geng et al. reported 98.82% overall accuracy for lane-changing behavior prediction on highD [23, 24]. These results are strong, but they mainly address the specific task of lane-change prediction. They do not directly solve the broader problem of learning latent human-driving regimes and predicting several maneuver types in one compact probabilistic model.

Deep learning has become a major direction in trajectory and maneuver prediction. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks were widely used to model temporal dependencies in driving sequences, while Graph Neural Networks (GNNs) were later introduced to better represent interactions between multiple traffic participants [25, 26]. More recently, Transformer-based models have shown strong performance by capturing long-range dependencies and multi-agent interactions through attention mechanisms [27–30]. Generative approaches such as Conditional Variational Autoencoder (CVAE) and diffusion models further support multimodal prediction by generating multiple plausible future trajectories under the same observed context [26, 31–34].

Although these methods are powerful, especially for continuous trajectory forecasting and lane-change-specific classification, most of them often operate as highly data-driven black-box models. For applications where uncertainty representation, temporal structure, and behavioral interpretability are important, this can be a limitation. However, the goal of this paper is different. The focus is not full trajectory generation, but probabilistic short-horizon prediction of discrete human-driver maneuvers with a model whose latent states can be inspected after training.

2.4. Positioning of This Work

The review above shows a trade-off between simplicity, predictive power, and interpretability. Rule-based models are transparent but limited in uncertainty representation. They depend strongly on hand-crafted assumptions. Learning-based models can achieve high prediction accuracy, especially for supervised

lane-change or trajectory forecasting tasks, but they often require large labeled datasets and careful tuning. Probabilistic temporal models provide a useful option when the aim is to model uncertainty and update predictions over time.

This paper uses a DBN because the target problem is short-horizon maneuver prediction from naturalistic highway trajectories, where semantic maneuver labels are not directly provided by the dataset. The proposed model learns latent behavior states from window-level motion and interaction features, then maps the learned joint states to semantic maneuver labels after training. This allows the model to perform online probabilistic prediction while keeping the learned behavior representation compact and inspectable. The work therefore focuses on structured human-driver maneuver prediction for HDVs, rather than full trajectory generation or downstream planning.

3. METHODOLOGY

This section describes the methodology used to construct and evaluate the proposed DBN-based maneuver prediction framework. It first presents the overall workflow and the selected highD data subset. It then explains the preprocessing steps, the sliding-window observation design, the DBN formulation, and the training and online prediction procedure. The full processing pipeline is summarized in Fig. 1.

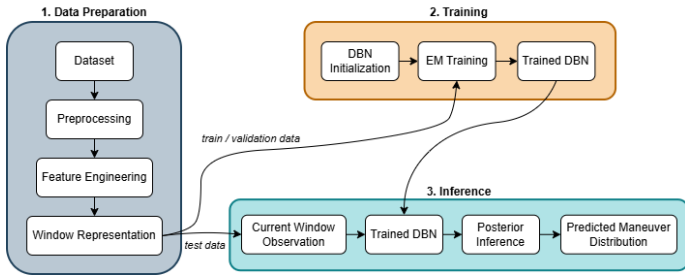


FIGURE 1: Overview of the proposed framework

3.1. Framework Overview

The proposed framework consists of three main stages: preprocessing of raw trajectory data, probabilistic temporal modeling, and causal online prediction. First, frame-level trajectories from highD are converted into a consistent representation that is independent of the original driving direction. Next, ego-motion, lane-related, interaction, and risk-related features are extracted and aggregated over short overlapping windows. Each window then forms one observation for the DBN, which models temporal evolution through two latent variables: a style state and an action state. After training, the model is used for online filtering and one-step-ahead maneuver prediction.

3.2. Dataset and Preprocessing

The method is evaluated on the highD dataset [7], which contains naturalistic highway trajectories recorded from drone videos on German highways. The full highD release contains 60 recordings collected at several highway locations, but this work uses only the first 25 recordings with three lanes in each driving

direction. This restriction keeps the road geometry and traffic setting more uniform across the selected subset, which supports more consistent behavior modeling while still leaving a large and temporally rich set of trajectories for window-based analysis. For each vehicle, the dataset provides frame-level position, velocity, acceleration, lane information, neighboring-vehicle identifiers, and traffic-safety indicators such as time headway (THW) and time to collision (TTC).

Because the selected recordings contain traffic moving in opposite roadway directions, the raw kinematic signals do not follow a consistent sign convention across all vehicles. A vehicle-centric normalization step is therefore applied so that the same maneuver is represented with the same sign convention regardless of the original travel direction. This produces a consistent frame-level representation for later feature extraction and windowization.

3.3. Data Analysis for Feature Design

Before defining the final observation vector, we performed an exploratory analysis of the selected highD subset to understand the behavior of the available signals and guide feature design. The analysis focused on frame-level kinematics, trajectory-level variability, and lane-change sparsity.

The main findings are that longitudinal acceleration and velocity carry broader dynamic ranges and clear class-dependent scale differences, while lateral variables are concentrated near zero but still show informative non-zero tails during lane changes. Trajectory-level variability also reveals that some vehicles remain nearly steady, whereas others exhibit much stronger longitudinal and lateral fluctuations over time. In addition, lane-change events are sparse relative to lane keeping, which means that a single binary lane-change indicator is not enough on its own and should be complemented by contextual and temporally aggregated features.

These observations motivate the final observation design used in this paper: the model combines longitudinal and lateral motion, lane geometry, interaction context, and risk-related signals, and it summarizes them over short overlapping windows instead of using individual frames directly.

3.4. Window-Based Observation Design

The observation vector combines ego-motion, lane-related, interaction, and risk-related information. It includes kinematic quantities such as longitudinal and lateral velocity and acceleration, lane-boundary distances, lane-change evidence, front-vehicle safety indicators, and selected relative features describing nearby traffic participants. These variables are designed to represent both the state of the ego vehicle and the surrounding traffic context relevant for maneuver prediction.

Rather than using individual frames directly, the model operates on short temporal windows. Each trajectory is converted into a sequence of overlapping windows with length $W = 50$ frames and stride $r = 10$ frames. Since the highD dataset is recorded at 25 Hz, each window covers 2.0 s of history and consecutive windows start every 0.4 s. Each window is summarized into one observation vector using simple temporal aggregation functions, which provide a compact representation of recent motion and interaction history while keeping the observation space manageable. The resulting windowization process is illustrated in Fig. 2.

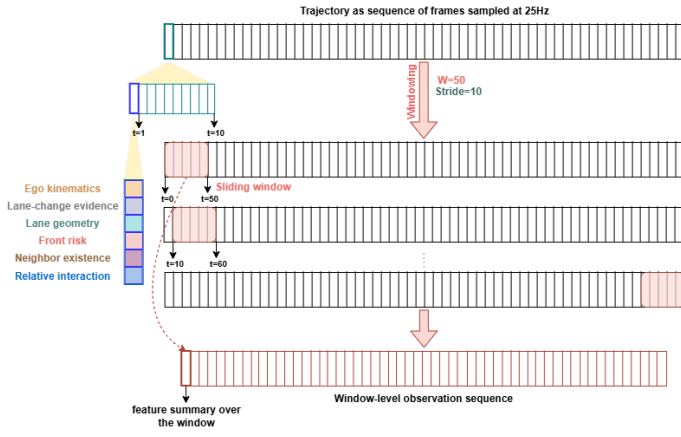


FIGURE 2: Illustration of the sliding-window representation

3.5. DBN Formulation

The proposed DBN uses two discrete latent variables at each time step: a style state S_t and an action state A_t . The style state captures a broader driving regime, while the action state represents the current maneuver. The observation vector O_t corresponds to the aggregated window-level features described above. The graphical structure in Fig. 3 shows the observation as a single node for compactness. In the implementation, this observation node is split into an action-related block and a style-related block.

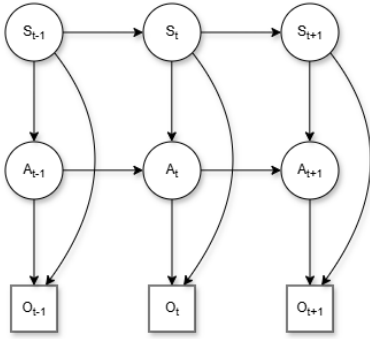


FIGURE 3: Structure of the proposed DBN

The temporal dependencies are factorized such that the style state evolves according to its previous value, while the action state depends on both the previous action and the current style. This structure reflects the assumption that the broader driving regime changes more slowly, whereas the short-term maneuver depends on both recent behavior and current traffic context.

The observation sequence for one vehicle is denoted by $O_{1:T} = (O_1, \dots, O_T)$, where each O_t is one window-level observation. In the emission model used in this paper, each observation is split into two feature blocks,

$$O_t = (O_t^{\text{act}}, O_t^{\text{sty}}). \quad (1)$$

The action-related block O_t^{act} contains ego-motion, lane-change, and lane-geometry features that describe the physical maneuver. The style-related block O_t^{sty} contains interaction, risk, and surrounding-traffic features that describe the driving context.

The joint distribution of the proposed DBN for a sequence of length T is written as

$$\begin{aligned} p(S_{1:T}, A_{1:T}, O_{1:T}^{\text{act}}, O_{1:T}^{\text{sty}}) \\ = p(S_1)p(A_1 | S_1)p(O_1^{\text{act}} | A_1)p(O_1^{\text{sty}} | S_1) \\ \times \prod_{t=2}^T p(S_t | S_{t-1})p(A_t | A_{t-1}, S_t) \\ \times p(O_t^{\text{act}} | A_t)p(O_t^{\text{sty}} | S_t). \end{aligned} \quad (2)$$

This factorization means that the action-related feature block is conditioned on the action state, while the style-related feature block is conditioned on the style state. The action state is still influenced by the current style state through the transition model $p(A_t | A_{t-1}, S_t)$.

The emission likelihood is therefore written as

$$p(O_t | S_t = s, A_t = a) = p(O_t^{\text{act}} | A_t = a)p(O_t^{\text{sty}} | S_t = s). \quad (3)$$

For each block $g \in \{\text{sty}, \text{act}\}$, continuous features are modeled with Gaussian distributions with diagonal covariance and binary features with Bernoulli distributions:

$$\begin{aligned} p(O_t^g | z_g) &= \prod_{d \in C_g} \mathcal{N}(o_{t,d}; \mu_{z_g,d}, \sigma_{z_g,d}^2) \\ &\times \prod_{b \in B_g} \text{Bern}(o_{t,b}; \rho_{z_g,b}), \end{aligned} \quad (4)$$

where $z_{\text{sty}} = s$ and $z_{\text{act}} = a$. The sets C_g and B_g denote the continuous and binary features in block g , respectively. The parameters $\mu_{z_g,d}$ and $\sigma_{z_g,d}^2$ define the Gaussian emission for continuous feature d , while $\rho_{z_g,b}$ defines the Bernoulli probability for binary feature b . This diagonal-covariance choice is a tractable emission-model approximation; it does not imply that the raw motion features are independent in the data.

In the experiments reported in this paper, the DBN uses two style states and four action states, denoted as S2-A4. This gives eight possible joint latent states. Each joint state is later mapped to a semantic maneuver label for interpretation and predictive evaluation.

3.6. Training and Online Prediction

Model parameters are estimated from the training data using the expectation-maximization (EM) algorithm. Continuous features are normalized using statistics computed from the training split only, and model initialization includes initial-state probabilities, transition probabilities, and emission parameters derived from the training data. The training procedure learns both the temporal dynamics of the latent states and the emission model linking latent states to window-level features.

After training, the model is used in a causal online setting. At time step t , the filtered belief over the latent state pair is updated

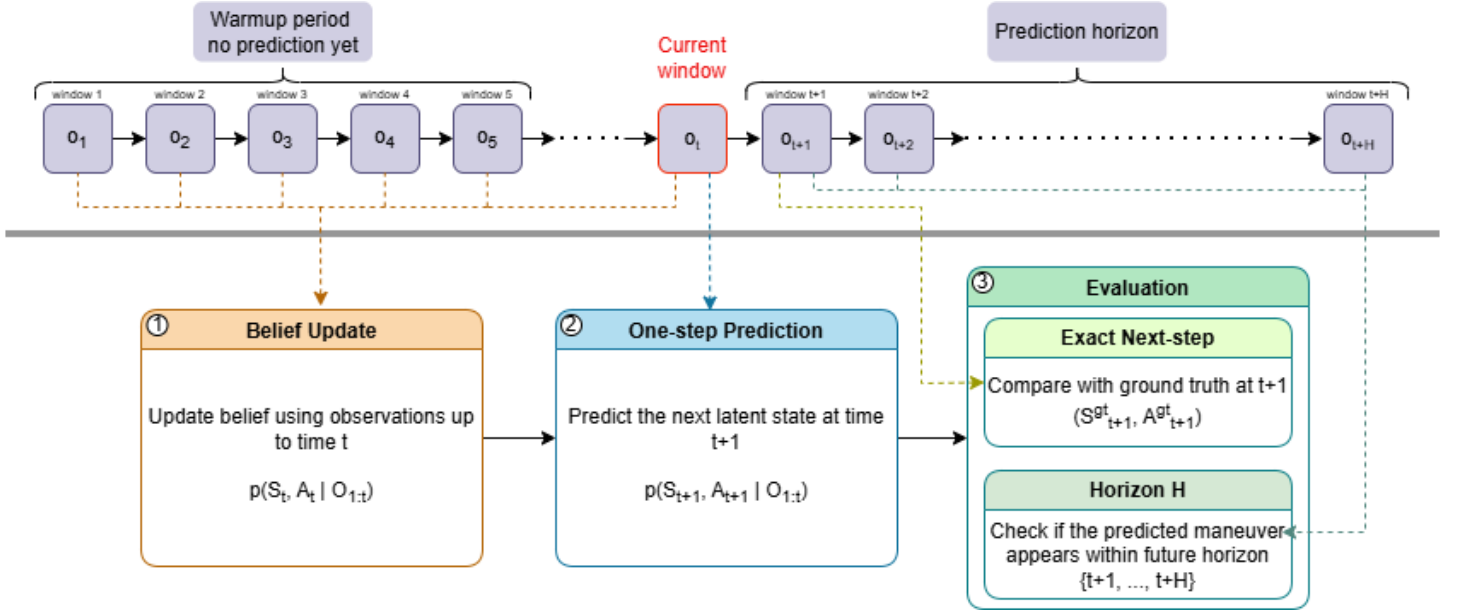


FIGURE 4: Causal online prediction pipeline used for one-step-ahead maneuver prediction

using the observations available up to that time, and the transition model is then used to obtain the one-step-ahead predictive distribution over future maneuvers. The causal prediction process is illustrated in Fig. 4.

After filtering, the one-step-ahead predictive distribution over the next joint latent state is obtained as

$$\begin{aligned}
 & p(S_{t+1} = s', A_{t+1} = a' | O_{1:t}) \\
 &= \sum_{s,a} p(S_t = s, A_t = a | O_{1:t}) p(S_{t+1} = s' | S_t = s) \quad (5) \\
 & \times p(A_{t+1} = a' | A_t = a, S_{t+1} = s').
 \end{aligned}$$

The predicted maneuver is then obtained from the most likely joint state and its semantic maneuver label.

4. EXPERIMENTAL SETUP

4.1. Data Split and Compared Configurations

The selected highD subset is partitioned at the vehicle level into disjoint training, validation, and test sets to prevent leakage across overlapping windows from the same trajectory. The split contains 35,318 training vehicles (70%), 5,045 validation vehicles (10%), and 10,092 test vehicles (20%). The training set is used for parameter estimation, the validation set is used for early stopping, and the test set is reserved for final evaluation.

Two S2-A4 DBN configurations are evaluated. Both models use the same latent structure with two style states and four action states, the same window representation, and the same train/validation/test split. The first model does not apply lane-change reweighting and is used as the main model. The second model, denoted as LC-B, applies lane-change-based weighting during EM training and is included as an imbalance-handling comparison.

Lane changes are relatively rare in naturalistic highway driving compared with lane keeping, cruising, braking, and acceleration. Therefore, maximum-likelihood training can be dominated

by more frequent non-lane-change windows. To study this effect, the LC-B variant gives windows with lane-change evidence a higher weight in the statistics used for parameter updates. This gives rare lateral maneuver evidence more influence during training, while keeping the online inference and test evaluation unchanged. Thus, LC-B is treated as a training-weighting variant, not as a separate DBN structure.

4.2. Evaluation Protocol

Evaluation is performed in two parts. First, held-out model fit is measured on complete unseen test sequences using likelihood-based metrics. This evaluates how well the trained DBN explains the observed window sequences. Second, online predictive evaluation is performed in a causal setting, where only observations up to the current time step are used to update the filtered belief and predict the next maneuver.

Because the DBN is trained without semantic maneuver labels, the learned joint latent states are interpreted after training through a two-stage procedure. First, for each joint state (s, a) , posterior-weighted feature statistics are computed from the original window-level features in the training set. This involves computing the posterior probability that each training window belongs to that joint state, then computing weighted means and standard deviations of the features, giving a profile of the typical behavior associated with that state. Second, each joint state is assigned a semantic maneuver label by applying domain-based rules to its feature profile. These rules use feature thresholds and decision logic based on the patterns of longitudinal motion, lateral motion, front-vehicle presence, and interaction context to classify windows into one of eight maneuver classes. The semantic maps are created separately for the two compared models, because each model can learn a different latent-state structure.

Since the highD dataset does not provide maneuver labels directly, reference labels for online evaluation are generated from

the preprocessed window features using a separate rule-based labeling procedure. These rules use cues such as longitudinal acceleration, lateral motion, lane-change evidence, and following behavior to assign each scored window to a maneuver class. The reference labels are used only for evaluation and are not used during EM training.

During online evaluation, the predicted label is obtained from the most likely DBN joint state through the corresponding semantic map and is compared with the rule-based reference label. Windows without a reliable reference label are excluded from scoring. These unclassified windows result from the rule set's limited coverage: they represent edge cases and boundary conditions that fall between the defined rule thresholds and thus cannot be reliably assigned to any single semantic class. The first five window observations of each test trajectory are used as a warmup period and are also excluded. After warmup, one-step-ahead predictions are generated sequentially. For horizon-based evaluation, the forecast horizon covers the next five window steps, corresponding to 2.0 s.

4.3. Evaluation Metrics

Model fit on unseen test sequences is assessed using the average negative log-likelihood (NLL) per time step. NLL measures how much probability the trained DBN assigns to the observed window sequences in the held-out test set. It is therefore used as a likelihood-based sanity check for the probabilistic model: a lower NLL indicates better held-out likelihood and helps verify that the model has learned the main structure of the data without only fitting the training set. However, NLL is not used as the main evidence for safe maneuver prediction. Because naturalistic highway data are dominated by common behaviors such as cruising and lane keeping, a model can obtain a good NLL while still making poor predictions for rare but important events such as lane changes. For this reason, NLL is reported only as a model-fit metric and is interpreted together with predictive metrics, class-wise results, and horizon-based evaluation.

The Bayesian Information Criterion (BIC) is also reported as a model-selection score that penalizes model complexity. Like NLL, BIC is used only to compare held-out model fit within the evaluated model family. Posterior occupancy diagnostics are used separately to inspect how effectively the learned latent states are used.

Online predictive performance is assessed using exact next-step accuracy, macro-averaged precision, recall, and F1-score. A horizon-based hit rate over 2.0 s is also reported. This metric counts a prediction as successful if the predicted semantic maneuver appears at least once within the next five window steps. It helps capture cases where a maneuver is predicted correctly but slightly shifted in time.

Average CPU latency is also measured for online belief update and next-step prediction. This is used to check whether the method can run faster than the 0.4 s interval between consecutive window observations.

5. RESULTS AND DISCUSSION

This section presents the results of the proposed DBN-based maneuver prediction framework. Both models use the same pre-

processing pipeline, vehicle-level split, window-based observation format, and online prediction protocol. The main goal of this comparison is to study whether the lane-change based weighting improves the final online prediction behavior.

5.1. Semantic Labels

For both models, the two style states are interpreted after training as low-interaction and high-interaction driving regimes. The low-interaction style represents driving with weaker front-vehicle influence, while the high-interaction style represents more constrained driving with stronger coupling to nearby traffic. The action-state meanings are model-specific because the latent states are learned in an unsupervised manner. Therefore, the semantic maps are kept separate for the two evaluated models, as shown in Table 1.

TABLE 1: Action semantics for the two models.

Action state	Main model	lc-B model
a_0	Cruising	Cruising
a_1	Braking	Braking
a_2	Acceleration with left lateral tendency	Acceleration
a_3	Acceleration with right lateral tendency	Lane change

5.2. Model Fit and Latent-State Usage

Table 2 compares the held-out model fit and latent-state usage of the two models. The main model gives the better held-out fit. Its NLL per step and BIC value are lower than those of the lc-B model. Here, $k_{\text{eff}}^{\text{joint}}$ denotes the effective number of occupied joint latent states, and $occ_{\text{max}}^{\text{joint}}$ denotes the maximum posterior occupancy of any joint state.

TABLE 2: Held-out fit and latent-state usage of the two models.

Model	NLL/step	BIC [M]	$k_{\text{eff}}^{\text{joint}}$	$occ_{\text{max}}^{\text{joint}}$
main	23.88	16.43	6.66	0.26
lc-B	30.94	21.29	3.94	0.36

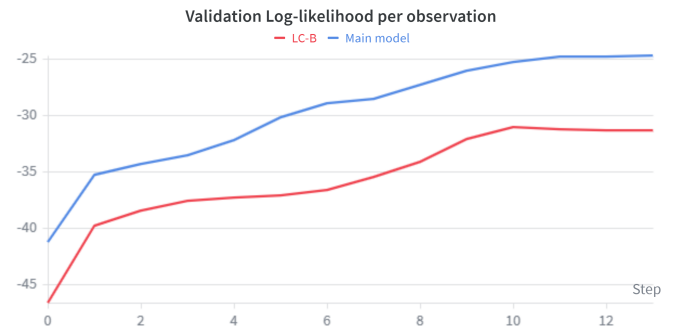


FIGURE 5: Validation curves during training for the two models.

Figure 5 shows the validation log-likelihood per observation during training. The main model consistently achieves higher validation likelihood than the lc-B model across the training steps and also converges to a better final value. This supports the held-out test results in Table 2, where the main model also shows better likelihood and lower BIC. Together, these results indicate that the main model captures the observation structure more effectively than the lc-B variant.

The occupancy values are used only as diagnostics. They should not be interpreted as a direct performance measure, because the maneuver classes are not equally frequent. For example, lane-change-related behavior is naturally rare, so a lane-change state is expected to have lower occupancy than cruising or braking states. The main point is therefore not that lc-B has lower occupancy by itself, but that this more concentrated state usage appears together with worse likelihood, worse BIC, and weaker online prediction results.

5.3. Online Prediction Performance

Table 3 summarizes the online prediction results. The main model gives the strongest prediction performance, which shows that the model is reliable for short-horizon joint style-action prediction.

TABLE 3: Online prediction performance.

Model	Accuracy	Precision	Recall	F1	Hit rate
main	79.4%	67.6%	82.0%	71.5%	81.3%
lc-B	64.2%	46.4%	40.2%	34.8%	67.5%

The lc-B model performs clearly worse. Its exact joint prediction accuracy drops to 64.2%, and its macro F1-score drops to 34.8%. The Hit rate also decreases to 67.5%. This means that the lc-B model does not only make more one-step errors, but also fails more often within the full 2 s forecast horizon.

The lower macro scores show that the lc-B degradation is not only caused by one dominant class. Instead, the prediction quality becomes weaker across several semantic classes. This is consistent with the model-fit results and with the more concentrated latent-state usage in Table 2. However, the lower effective joint-state usage should be interpreted carefully, because rare behaviors such as lane changes are not expected to occupy the same amount of data as common behaviors.

Figure 6 shows the confusion matrix for the main model. The diagonal structure is clear, which means that the model predicts most joint states correctly. The strongest classes are high-interaction braking, high-interaction cruising, and the two high-interaction acceleration-with-lateral-tendency states.

Table 4 gives the class-wise precision and recall for the main model. The two cruising classes have high recall, which means that most cruising samples are recovered correctly. However, their precision values are lower, which means that some braking and lateral-tendency samples are also predicted as cruising. This is reasonable because weak braking or early lateral motion can look similar to cruising within a short observation window.

The high-interaction braking class gives the strongest precision, while both braking classes show strong recall. The

TABLE 4: Class-wise precision and recall for the main model.

Class	Precision	Recall
LI/cruising	42.9%	95.7%
LI/braking	65.7%	79.9%
LI/accel. with left lateral tendency	60.3%	82.4%
LI/accel. with right lateral tendency	62.6%	73.7%
HI/cruising	40.1%	95.3%
HI/braking	99.7%	81.0%
HI/accel. with left lateral tendency	81.3%	79.7%
HI/accel. with right lateral tendency	88.3%	68.5%

two acceleration-with-lateral-tendency classes are also predicted well, although the high-interaction right-lateral-tendency class has lower recall.

The lc-B confusion matrix is shown in Appendix A. Compared with the main model, the diagonal pattern is weaker. The lc-B model predicts many samples as high-interaction cruising, high-interaction braking, or high-interaction acceleration. Low-interaction classes are often shifted into the corresponding high-interaction classes.

The strongest lc-B classes are high-interaction braking and high-interaction acceleration. These classes have high recall and contain many samples. However, low-interaction cruising, low-interaction braking, and low-interaction acceleration have weak recall. The lane-change classes are also difficult. In particular, many high-interaction lane-change samples are predicted as high-interaction acceleration or high-interaction braking. This means that the lc-B model does not produce a clean lane-change separation, even though the semantic map gives special attention to lane-change evidence.

This result suggests that the lc-B weighting does not improve the learned representation in the present setting. Lane-change events are naturally much rarer than cruising, braking, or acceleration, so a lower occupancy of lane-change-related states is not by itself unexpected. However, even with additional lane-change emphasis, the lc-B model does not achieve cleaner class separation or better online prediction performance. Instead, its predictive scores decrease and its latent-state usage becomes more concentrated. This indicates that the added weighting does not translate into a more useful model for short-horizon maneuver prediction in this dataset.

The exact one-step metric is strict because the predicted class must match the next window exactly. For a driving application, a prediction can still be useful if the correct behavior appears slightly later. Therefore, both models are also evaluated with a 2 s horizon, which corresponds to the next five future windows.

The main model reaches 91,624 hits out of 112,643 scored cases, giving a Hit rate of 81.3%. Most of these hits occur at the first future step. The mean time-to-hit is 0.42 s, which shows that when the main model is correct, the correct behavior usually appears almost immediately.

The lc-B model reaches 88,405 hits out of 130,891 scored cases, giving a Hit rate of 67.5%. Its mean time-to-hit is 0.45 s. Thus, lc-B does not mainly fail because correct predictions occur too late. Instead, it produces fewer correct predictions overall.

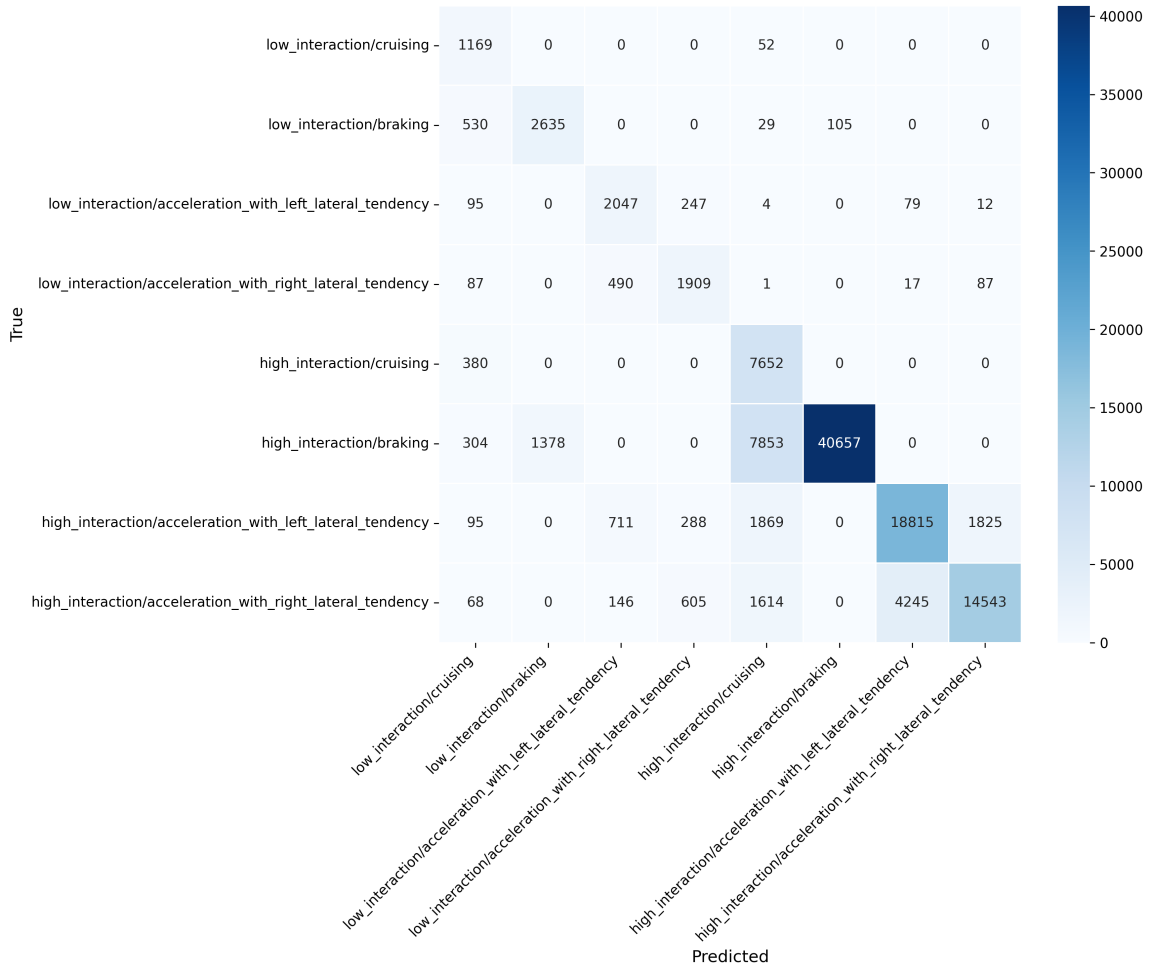


FIGURE 6: Confusion matrix for the main model.

The horizon results support the same conclusion as the one-step metrics. The main model is more reliable, while lc-B reduces the number of successful forecasts. For both models, most successful predictions occur within the first horizon step. This means that the DBN is mainly useful for immediate short-horizon maneuver prediction.

5.4. Computational Efficiency

The online prediction step is computationally light. The input data are sampled at 25 Hz and a new window is generated every 10 frames. Therefore, the available prediction interval is 0.4 s, or 400 ms.

The latency benchmark was carried out on a CPU-only setup using an AMD Ryzen 9 7950X 16-Core processor. In the benchmark, the CPU runtime for the window-to-prediction step is about 0.19 ms for one vehicle. The end-to-end CPU runtime, including frame-to-window construction and prediction, is about 0.61 ms for one vehicle. For five vehicles processed in batched mode, the end-to-end CPU runtime is about 1.97 ms. All of these values are far below the 400 ms interval.

These results show that the proposed DBN can run in real time on standard CPU hardware. Even the end-to-end latency remains much smaller than the available 400 ms update interval.

TABLE 5: Mean online latency for the prediction pipeline on CPU.

Scenario	Pipeline	Mean time
Single vehicle	window→predict	0.19 ms
Single vehicle	frame→window→predict	0.61 ms
Batched, 5 vehicles	frame→window→predict	1.97 ms

5.5. Discussion

The results show that the main model gives the best overall behavior prediction performance. It has better held-out fit, better latent-state usage, higher one-step accuracy, higher macro F1-score, and higher Hit rate than the lc-B model.

The comparison also shows that stronger lane-change weighting is not automatically useful. Although lane changes are important, lc-B makes the learned representation less balanced. The model uses fewer effective joint states and places most probability mass in one style state. This weakens the separation between low-interaction and high-interaction regimes. It also makes the online prediction less stable across classes.

The main model still has limitations. Some errors remain between behaviors that are close to each other, such as cruising

and weak braking. Overall, the main S2A4 DBN provides the best trade-off between prediction accuracy, latent-state balance, and real-time feasibility. It is therefore used as the final model for the proposed short-horizon human-driver maneuver prediction framework.

6. CONCLUSION

This paper presented a DBN-based framework for probabilistic short-horizon maneuver prediction of human-driven vehicles in highway traffic. The model uses window-based trajectory observations and predicts joint style-action states in a causal online setting. The style state represents the broader interaction regime, while the action state represents the current maneuver tendency.

The results show that the main model without lane-change weighting gives the best overall performance. It achieved a joint one-step accuracy of 79.4%, a macro F1-score of 71.5%, and a Hit rate value of 81.3%. The lc-B model performed worse, with 64.2% accuracy, 34.8% macro F1-score, and 67.5% Hit rate. This indicates that stronger lane-change weighting did not improve the final prediction behavior in this setting. The main model also satisfies the real-time requirement, with an end-to-end CPU runtime of about 0.61 ms for one vehicle and 1.97 ms for five vehicles in batched mode, both far below the 400 ms prediction interval.

The main limitation is that some semantically close behaviors, such as cruising and weak braking or the two lateral-tendency acceleration states, remain difficult to separate. Future work should therefore improve the observation design, test the method on more diverse datasets, and study alternative latent-state structures. The probabilistic predictions can also be integrated into a larger automated-driving or AV decision-making pipeline to support risk assessment and safe interaction with human drivers.

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APPENDIX A. CONFUSION MATRIX FOR LC-B

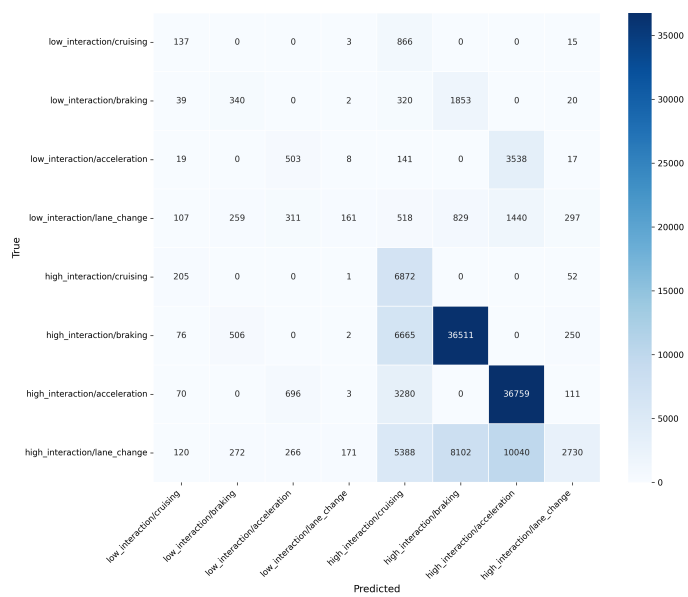


FIGURE 7: Confusion matrix for the lc-B model.